

CGA: Conditioning as Group Action

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Abstract

Condition a network by rotating its features, with rotation angles linear in the condition, and combined conditions compose automatically: rotations add, so the network handles combinations it never saw in training. FiLM and hypernetwork conditioners fail on the same combinations at up to $43\times$ the cost. We present **CGA** (Conditioning as Group Action), a framework that treats feature conditioning $y = T(c)x + \beta(c)$ as a group representation of the condition space: the condition c enters the operator linearly in an abelian Lie algebra, $T(c) = P \exp(A(Wc))P^\top$, so the homomorphism law $T(c_1+c_2) = T(c_1)T(c_2)$ holds exactly, for any weights, by construction. Choosing the algebra recovers deployed mechanisms as special cases. The diagonal algebra yields an exactly compositional variant of FiLM; FiLM’s usual MLP head parametrizes the same orbit without the homomorphism, and our ablations show this missing law is what fails under composition. The maximal torus $\mathfrak{so}(2)^{d/2}$ with scalar $c=n$ recovers RoPE, and with learned generators the multiplicative GRAPE construction, transplanted from attention positions to arbitrary condition vectors in FiLM’s role. Our instantiation, planar rotations with angles Wc conjugated by a Group-and-Shuffle structured orthogonal basis, is exactly orthogonal, equals the identity at $c=0$ and at initialization, and costs $1.11\times$ FiLM’s conditioning FLOPs; a dense learned basis provably exceeds any FiLM-comparable budget, and transformer sites need no explicit basis at all, as in RoPE. Under a pre-registered, budget-fair protocol against FiLM, conditional LayerNorm, concatenation, and two input-conditioned hypernetworks (up to $48\times$ our parameters), CGA achieves the lowest compositional OOD error on all six suites we evaluate, with complete seed separation throughout ($p=9.1\times 10^{-5}$, $N=10$), including zero-shot extrapolation to compositions longer than any seen in training. Swapping only the linear map for an MLP degrades OOD error by up to 4.3×10^4 . The most expressive baselines generalize worst. Two pre-registered gates returned negatives that bound the method: a $1.46\times$ in-distribution fit penalty on rich scenes, and a content-conditioning failure in diffusion; we report both as registered. Exact orthogonality trades fit for guarantees, and we quantify the trade.

1 The framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM [1] computes $y = \gamma(c) \odot x + \beta(c)$; adaLN in DiT blocks [2] is its normalized form; hypernetworks emit $W(c)$ [3]. All are instances of

$$y = T(c)x + \beta(c), \tag{1}$$

and they differ only in the family from which the per-sample operator $T(c)$ is drawn.

Definition 1 (Compositional conditioner). $T : \mathbb{R}^k \rightarrow \text{GL}(d)$ is compositional if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1+c_2) = T(c_1)T(c_2)$ and $T(0) = I$.

Proposition 1 (Exact composition by construction). Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $A : \mathbb{R}^m \rightarrow \mathfrak{a}$ linear, $W \in \mathbb{R}^{m \times k}$, and P orthogonal. Then $T(c) = P \exp(A(Wc))P^\top$ is a compositional conditioner for every W, P ; if $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.

Proof. $\exp(X)\exp(Y) = \exp(X+Y)$ for commuting X, Y ; $A \circ W$ is linear; conjugation preserves products and $\exp(0) = I$. $\square$

Composition is a property of the parametrization: it holds at initialization, during training, and off-distribution. The learnable parts, W and P , select which one-parameter subgroups the condition drives and how fast.

Table 1: Compositional OOD error (mean over 10 seeds). CGA is lowest in every column with complete seed separation vs. the strongest hypernetwork baseline ($p=9.1\times 10^{-5}$, $\delta=-1.0$). S1–S3b pass the full pre-registered gate; S4 returns a registered negative on a separate in-distribution criterion (text).

Conditioner	S1	S2	S3	S3b	S4
FiLM	1.3×10^{-2}	1.6×10^{-2}	3.0×10^{-3}	5.9×10^{-3}	4.5×10^{-3}
Concat-MLP	1.5×10^{-2}	1.1×10^{-2}	2.9×10^{-3}	6.3×10^{-3}	5.4×10^{-3}
Cond-LN	2.4×10^{-2}	2.7×10^{-2}	3.2×10^{-3}	6.1×10^{-3}	8.6×10^{-3}
Hypernetwork	1.9×10^{-3}	3.4×10^{-3}	3.8×10^{-3}	7.0×10^{-3}	9.3×10^{-3}
Dyn. low-rank	5.7×10^{-3}	6.1×10^{-3}	5.1×10^{-3}	8.1×10^{-3}	2.0×10^{-2}
MLP-head (abl.)	–	1.1×10^{-3}	2.7×10^{-3}	5.0×10^{-3}	3.3×10^{-3}
Lie (ours)	7.4×10^{-4}	3.3×10^{-8}	1.8×10^{-3}	4.0×10^{-3}	1.2×10^{-3}

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms exactly:

$\mathfrak{a}$	condition	P	recovered mechanism
$\text{diag}(\mathbb{R}^d)$	c	I	<i>exponential FiLM</i> : $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	this work (FiLM’s role)

Standard FiLM parametrizes the same diagonal orbit through an MLP head $\gamma(c)$: the orbit matches, the homomorphism is absent, and §2 shows this missing law is what fails under composition. Our ablation with an MLP angle head (same orbit, same basis, more compute) loses by up to 4.3×10^4 . The PEFT literature (LoRA, OFT, BOFT, HRA, Group-and-Shuffle [6–10]) owns these operator families as fixed weight adaptations; CGA places them in FiLM’s role, per-sample and in activation space, where the composition question exists.

Cost, and where the basis comes from. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs and no matrix exponential. The basis carries the entire budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is $1.52\times$ FiLM’s whole conditioning path at $d=128$ and exceeds any FiLM-comparable ceiling. Two resolutions work. (i) A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to $1.11\times$ FiLM; P is undercomplete (4,800 parameters vs. $\dim \text{SO}(128)=8,128$) yet suffices, because P need only expose the planes the condition acts on. (ii) Transformer sites need no explicit P : the adjacent dense projections absorb the basis, which is how RoPE deploys. The operator satisfies identity initialization ($W=0 \Rightarrow T \equiv I$), $T(0) = I$ structurally, and $\sigma(T(c)) \equiv 1$; unit tests pin all three invariants in the release.

2 Experiments

Protocol. Every suite is pre-registered: margins, tests ($N=10$ seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff’s $|\delta| \geq 0.474$), and single-read OOD splits are fixed before each run. All arms share identical backbones. A shared counter enforces conditioning cost $\leq 1.20\times$ FiLM and parameters $\leq 1.05\times$ the smallest hypernetwork baseline; baselines may be larger, and reach $48\times$. We report every verdict as registered, including the negatives. Baselines: FiLM, conditional LayerNorm, Concat-MLP, dense hypernetwork $I+\Delta W(c)$, dynamic low-rank $I+A(c)B(c)^\top$, plus an MLP-head ablation (identical basis, more compute, no linearity).

Suites. **S1/S2 (synthetic):** learn $y = M(c)x$ for multi-hot compositions of 8 rotation primitives; S2 hides the primitives’ planes behind a random basis B and additionally tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** conditional image transformation. The model encodes a real image, applies $T(\Delta)$ on the learned latent for a factor-change Δ , and decodes; pixel MSE against the dataset’s unique ground-truth image scores the result. OOD means never-trained two-factor change types; S3b/S4 include a categorical (shape) factor encoded as a one-hot difference.

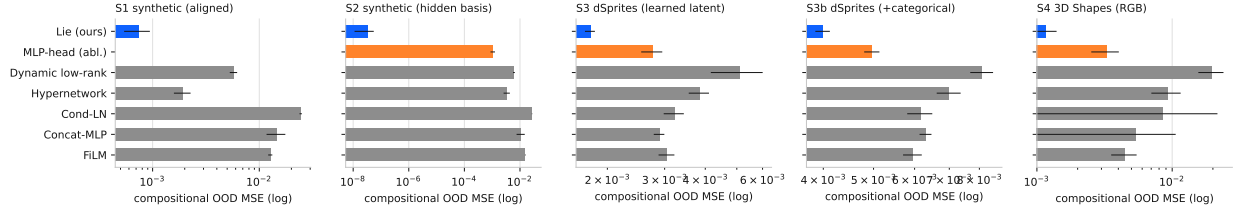


Figure 1: OOD error by conditioner (log scale; blue = CGA, orange = MLP-head ablation).

Findings. **(1) Group-structured conditions: solved.** On S2, CGA reaches 3.3×10^{-8} OOD MSE, five orders below the best baseline, and 5.3×10^{-8} on never-trained triples: error stays flat as composition length grows. Learned angles recover the true generators to 2×10^{-4} rad. A coordinate-aligned ablation ($P=I$) collapses to FiLM’s error, so basis discovery carries the result. **(2) The advantage survives learned latents.** On dSprites every arm fits in-distribution identically, which isolates composition: CGA’s OOD/in-dist gap is $1.25 \times$ ($1.63 \times$ with the categorical factor) vs. 2.1 – $3.3 \times$ for every baseline, a 53.8% (42.9%) margin over the strongest baseline at $39 \times$ fewer conditioning FLOPs. **(3) Linearity is the mechanism.** The MLP-head ablation shares the basis and the orbit and spends more compute; it loses by $4.3 \times 10^4 \times$ on synthetic triples and $1.55 \times$ on dSprites. **(4) Capacity inverts.** The two hypernetwork arms can represent any operator, and they post the worst compositional gaps on images (2.7 – $26 \times$): the extra expressivity memorizes training condition types. **(5) The registered negatives.** On 3D Shapes, CGA posts the best OOD error (87.5% margin; gap $1.13 \times$ vs. 13 – $26 \times$) and pays a $1.46 \times$ in-distribution fit penalty that trips the pre-registered no-regression gate ($\leq 1.10 \times$). In a conditional-diffusion suite, rotation conditioning underfits when the condition must specify image content: a rotation preserves information, so it transforms representations well and injects content poorly. We report both suites as negatives, as registered. Exact orthogonality trades fit for guarantees; our protocol prices the trade.

3 Discussion

CGA does for conditioning what RoPE and GRAPE did for position: it moves composition from a behavior the network must learn into a law the parametrization guarantees. The evidence here is deliberately controlled (64×64 images, known factors); the deployment experiment is the adaLN site of diffusion transformers, where the basis-free variant drops in below FiLM’s marginal cost. Three boundaries are already quantified: categorical factors attenuate the advantage, richer scenes price the orthogonality constraint in in-distribution fit, and content specification defeats a transformation channel. These motivate the abelian-algebra generalizations of Proposition 1 beyond the pure torus (torus $\times$ bounded diagonal) as the next design point. Adjacent work learns group actions on latents for world models [11, 12]; CGA lives in FiLM’s role under budget parity, and its composition law holds by construction instead of by training objective.

Reproducibility. Every suite is pre-registered; every number regenerates from append-only logs; unit tests pin the operator invariants and the fairness counters; the repository discloses all amendments and one accounting erratum.

References

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